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SYSTEM REQUIREMENTS SPECIFICATIONS

GAMIFIED FINANCIAL VISUALIZER

For

Demo 3

Presented by :



CodeBlooded

MENU

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Introduction

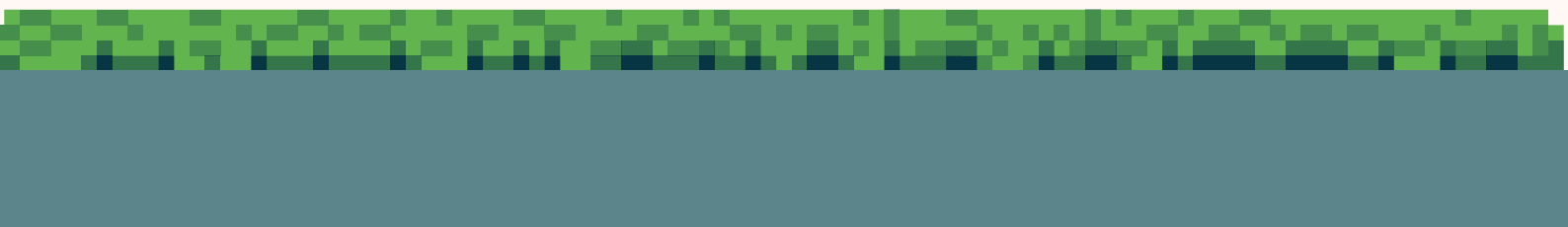
This System Requirements Specification (SRS) outlines the structure and behaviour of the Gamified Financial Visualizer, a cross-platform application designed to promote financial literacy through gamification, AI-powered insights, and augmented reality. The system enables users to manage their finances in an engaging and educational environment.

The purpose of this document is to define the system's functional and non-functional requirements, providing a shared reference for stakeholders, including developers, testers, and academic supervisors. It ensures a clear understanding of the system's objectives, scope, and constraints.

Included in this SRS are:

- Business Needs and Project Scope
- User Characteristics and User Stories
- Functional Requirements and Use Cases
- System Requirements (functional and non-functional)
- Traceability Matrix

This document will support the system's development lifecycle and evolve as the project progresses.



Business Needs

The Gamified Financial Visualizer addresses a growing need for accessible, engaging, and educational financial tools, particularly among younger users and students with limited financial literacy. Traditional finance apps often lack interactivity, personalization, and user motivation, which leads to disengagement and poor financial habits.

This project aims to solve that by integrating gamification, augmented reality (AR), and artificial intelligence (AI) into a single platform that transforms how users interact with their financial data.

The key business needs driving the system are:

Financial Literacy Promotion

To help users build strong financial habits by making financial education engaging, interactive, and rewarding.

User Engagement & Retention

To motivate continued use through gamified elements, AR visualizations, and reward-based learning, increasing user retention and satisfaction.

Personalized Financial Insights

To provide users with AI-driven, tailored financial feedback and suggestions based on their transaction data and goals.

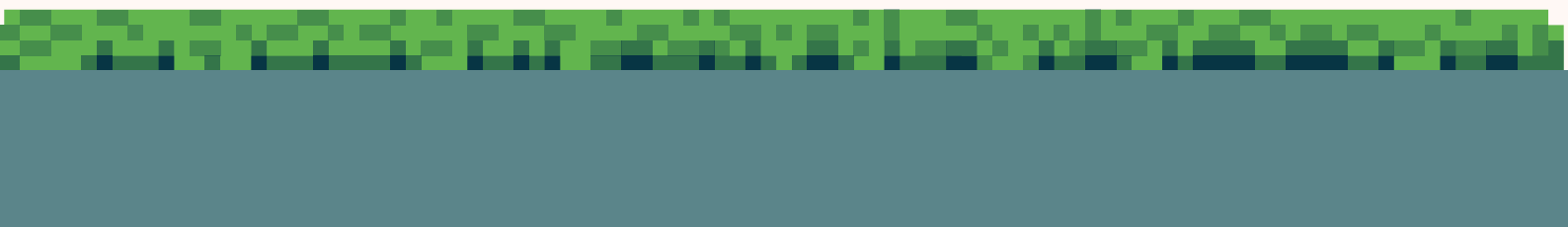
Cross-Platform Accessibility

To support both mobile and web platforms, ensuring users can manage and visualize their finances anytime, anywhere.

Security & Trust

To ensure user confidence through strong data protection, privacy practices, and compliance with legal standards (e.g., GDPR, POPIA).

By meeting these needs, the system supports its core objective: to transform financial management into an engaging, educational, and user-friendly experience.



Project Scope

The Gamified Financial Visualizer is a cross-platform application designed to improve financial literacy and management through interactive visualizations, gamified learning, and AI-driven financial insights. This project focuses on developing a functional platform that demonstrates core features while remaining feasible within academic timelines and technical constraints.

In Scope:

Financial Data Visualization

Users can view and interact with their financial data through dashboards, 3D charts, and AR scenes.

Transaction Categorization

AI models (e.g., DistilBERT) will classify and label transactions based on user input and historical data.

Personalized Financial Advice

Users receive context-aware tips, budgeting suggestions, and progress tracking from a local AI assistant.

Gamified Learning Modules

Educational content is delivered in a quiz-based, point-earning format to encourage consistent financial learning.

Augmented Reality Features

Users can visualize their financial goals, progress, and insights using AR experiences powered by Unity.

Cross-Platform Support

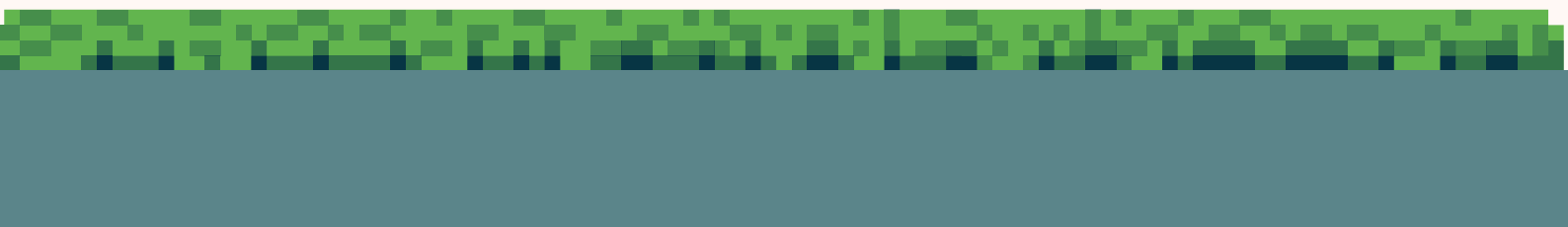
The application will run on mobile (React Native) and web (React.js), with shared components for consistent behaviour.

Secure Authentication and Privacy Controls

Implementation of secure login, encryption, and data anonymization to comply with privacy regulations.

Community and Social support

Users can invite and view friends to join in shared challenges and earn points together.



User Characteristics

User Type	Description
Registered User	Authenticated users with access to all features including tracking, visualization, education, and community.
Gamification-Oriented User	Users who are motivated by points, badges, and dynamic AR-based financial visualizations.
Financial Learner	Users who engage with the educational modules and AI assistant to enhance their financial literacy.

User Stories

User Management

1. **As a new user**, I want to register using my email, full name, and username so that I can access the platform.
 - a. Acceptance Criteria:
 - i. Given I am on the registration page, when I enter valid details, then my account should be created.
2. **As a user**, I want to log in securely using my username and password so that I can access my account.
 - a. Acceptance Criteria:
 - i. Given I have an account, when I enter correct credentials, then I should be granted access.
3. **As a user**, I want to recover my password via email so that I can regain access to my account.
 - a. Acceptance Criteria:
 - i. Given I click 'forgot password', when I submit my email, then I should receive a recovery link
4. **As a user**, I want to enable 2FA so that my account has an extra layer of security.
 - a. Acceptance Criteria:
 - i. Given I have enabled 2FA, when I log in, then I should be prompted for a code sent to my email.

Financial Asset Visualization (AR)

1. **As a user**, I want to view my financial data as a 3D city in AR so that I can understand my finances visually.
 - a. Acceptance Criteria:
 - i. Given I launch AR view, when financial data is loaded, then I should see it as a city with financial data tooltips.
2. **As a user**, I want my AR city to update in real-time so that changes reflect immediately.
 - a. Acceptance Criteria:

- i. Given I input a new transaction, when the AR view is active, then the visual should update instantly.

3. **As a user**, I want to compare 3D and AR views so that I can understand financial data from multiple perspectives.

a. Acceptance Criteria:

- i. Given I switch views, when I toggle from 3D to AR, then the same data should be shown in both.

Expense and Income Tracking

1. **As a user**, I want the system to auto-classify transactions from my bank statements so that I don't have to do it manually.

a. Acceptance Criteria:

- i. Given I upload a PDF, when the AI processes it, then my expenses are categorized automatically.

2. **As a user**, I want to manually input and categorize transactions so I can track them independently.

a. Acceptance Criteria:

- i. Given I open the form, when I input data, then it should be saved and categorized.

3. **As a user**, I want to edit or delete transactions so that I can maintain accurate records.

a. Acceptance Criteria:

- i. Given I view a transaction, when I click edit/delete, then changes should reflect in my records.

4. **As a user**, I want to be able to sort and filter using date, transaction type or amount

a. Acceptance Criteria:

- i. Given I click on the sort or filter button and select a criteria, the transactions table updates automatically.

5. **As a user**, I want to define my own categories so that I can organize finances in my way.

a. Acceptance Criteria:

- i. Given I access category settings, when I create a new one, then it appears in my options.

6. **As a user**, I want to create and track monthly budgets link to transactions categories.

a. Acceptance Criteria:

i. Given I create a budget, I can link transactions to the budget to track my spending.

7. **As a user**, I want to create manual accounts so I can simulate offline assets.

a. Acceptance Criteria:

i. Given I open account creation, when I input data, then it appears in my list of accounts.

8. **As a user**, I want a insights dashboard to visualize my finances so I can understand trends and breakdowns.

a. Acceptance Criteria:

i. Given I access the dashboard, when I have data logged, then graphs and summaries are shown.

9. **As a user**, I want my spending to be analysed and scored so I can improve my financial behaviour.

a. Acceptance Criteria:

i. Given I log new expenses, when the machine learning model runs an analysis, then a score is shown in my dashboard.

10. **As a user**, I want to search for transactions using keywords or prompts so I can find them quickly.

a. Acceptance Criteria:

i. Given I enter a search query, when results match, then they are displayed.

Educational Modules

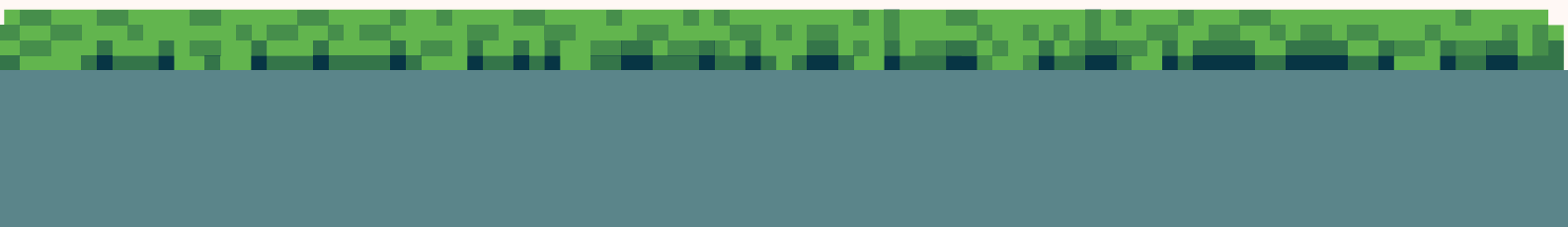
1. **As a user**, I want to view educational modules so I can improve financial literacy.

a. Acceptance Criteria:

i. Given I am on the education dashboard, when I click a module, then it opens with its content.

2. **As a user**, I want to get AI-based financial advice so I can learn from intelligent feedback.

a. Acceptance Criteria:



i. Given I type a question, when I click submit, then I get a personalized answer.

3. **As a user**, I want to earn rewards for completing quizzes, so I stay motivated.

a. Acceptance Criteria:

i. Given I complete a module and pass the quiz, then I earn points and feedback.

Goal Management

1. **As a user**, I want to create savings goals so I can track toward big purchases.

a. Acceptance Criteria:

i. Given I fill out the goal form, when I click save, then the goal is tracked.

2. **As a user**, I want to see visual progress toward my goals, so I stay motivated.

a. Acceptance Criteria:

i. Given I log progress, when I view the goal, then a visual bar updates accordingly.

3. **As a user**, I want notifications for approaching or completed goals, so I don't miss them.

a. Acceptance Criteria:

i. Given progress is 90% or 100%, when I open the app, then a message or popup appears.

4. **As a user**, I want AI to suggest savings tips so I can reach goals faster.

a. Acceptance Criteria:

i. Given AI sees overspending, when it runs analysis, then it provides recommendations.

Community Interaction

1. **As a user**, I want to find and add friends by username so I can build my network.
 - a. Acceptance Criteria:
 - i. Given I search a username, when results show, then I can send an invite.
2. **As a user**, I want to be able to remove friends from my lists
 - a. Acceptance Criteria:
 - i. Given I interact on a current friend, I click on 'remove friend', the user is removed
3. **As a user**, I want to join community challenges, so I stay consistent through competition.
 - a. Acceptance Criteria:
 - i. Given I start or join a challenge in a joined community, when I make progress, then it reflects in real-time.
4. **As a user**, I want to share achievements so I can celebrate progress.
 - a. Acceptance Criteria:
 - i. Given I complete a milestone, when I click 'Share', then a visual badge is generated.
5. **As a user**, I want to see a leaderboard so I can track how I rank among peers.
 - o Acceptance Criteria:
 - i. Given I open the leaderboard, when it's loaded, then I see my score and friends' scores.
6. **As a user**, I want my friends to get notified when I achieve goals, so I get recognition.
 - o Acceptance Criteria:
 - i. Given I complete a milestone, when it's logged, then friends are notified.
7. **As a user**, I want to be able to create a community for friends to join to participate in challenges together.
 - o Acceptance Criteria:
 - i. Given I complete the create form for the community, I can invite friends and set a banner for the community.
8. **As a user**, I want to be able to like and comment on community posts
 - o Acceptance Criteria:
 - i. Given I see a post, and I like and comment to the posts, the like count increments and the comment appears below the post.

Personalization

1. **As a user**, I want to personalize my AR world, so it reflects my preferences.
 - a. Acceptance Criteria:
 - i. Given I have enough points, when I unlock a skin, then it is applied to my AR interface.
2. **As a user**, I want to set notification and AI preferences, so the system adapts to my needs.
 - a. Acceptance Criteria:
 - i. Given I access settings, when I change preferences, then they are saved and used.

Support

1. **As a user**, I want to view to the help page related to the subsystem to receive guidance on how to use the page.
 - o Acceptance Criteria:
 - i. Given I click on the help, it will take to me the help page for that certain page and display information.
2. **As I user**, I want to be able to view video and image tutorials on how to use features on the system.
 - o Acceptance Criteria:
 - i. Given I click on the tutorial for a use case, I have access to the image slides and the video tutorial.

Functional Requirements

R1. User Management (Subsystem: Authentication & User Profile)

R1.1. User Registration and Login

- R1.1.1. Users shall register using their email, full name, and a chosen username.
- R1.1.2. Users shall securely log in using a username and password.
- R1.1.3. The system shall validate login credentials against securely stored hashes.

R1.2. Account Security

- R1.2.1. The system shall support two-factor authentication via email.
- R1.2.2. The system shall allow password recovery via email verification.

R1.3. Profile Management

- R1.3.1. Users shall be able to update personal information for personalization like avatars.
 - R1.3.2. The system shall allow users to delete their account.
-

R2. Financial Asset Visualization (Subsystem: AR Visualization Engine)

R2.1. Augmented Reality Financial Worlds

- R2.1.1. The system shall render an AR environment showing virtual 3D city based on financial activity.
- R2.1.2. Users shall be able view tooltips on the 3D city to view necessary insights on each building.
- R2.1.3. Assets shall update in real-time with new data inputs.

R2.2. Personalization

- R2.2.1. The system shall allow users to be able to switch between unlocked themes which will change in real-time
 - R2.2.2. The system shall highlight financial growth and risk areas visually.
-

R3. Expense and Income Tracking (Subsystem: Finance Manager + AI Classification)

R3.1. Automatic Classification with AI

- R3.1.1. Users shall upload bank statement PDFs for automatic categorization.
- R3.1.2. The AI shall classify transactions by type.

R3.2. Manual Financial Entry

- R3.2.1. Users shall manually input incomes and expenses.
- R3.2.2. Users shall edit or delete entries post submission.
- R3.2.3. Users shall define custom categories and associate transactions.

R3.3. Habit & Reminder Features

- R3.3.1. The system shall send reminders to log transactions.
- R3.3.2. The system shall support recurring entries for debit orders and interest.

R3.4. Financial Budgeting

- R3.4.1. Users will be able to create budgets on transactions categories.
- R3.4.2. The system shall notify the user when they maximise their budget.

R3.4. Insights dashboards & Analytics

- R3.4.1. An interactive dashboard shall display graphs for categories, dates, etc.
 - R3.4.2. Users shall view point scores representing financial health and gamified performance.
 - R3.4.3. System shall be able to analyze spending habits and provide scoring insights and trends of spending to the user.
 - R3.4.4. Users shall be able to prompt the system of AI insights regarding the spending to gain insights and information.
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R4. Community Interaction (Subsystem: Social Layer)

R4.1. Social Features

- R4.1.1. Users shall invite friends using usernames.
- R4.1.2. Users shall compare spending and gamified progress with others.
- R4.1.3. The system shall support leaderboards.
- R4.1.4. Friends shall receive notifications when a user earns rewards.
- R4.1.4. Users shall be able to create communities whereby friends can be sent invites to.

R4.2. Gamified Social Engagement

- R4.2.1. The system shall facilitate challenges to communities created by community members.
 - R4.2.2. Users shall share achievements posts within the social page.
 - R4.2.3. The system shall facilitate a shared point system that will allow all members of the community to gain points on a completed challenge.
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R5. Goal Managements (Subsystem: Financial Goal Engine)

R5.1. Goal Creation & Progression

- R5.1.1. Users shall set goals with target dates and target amounts.
- R5.1.2. The system shall track and visualize goal progress.
- R5.1.3. Notifications shall alert users near goal deadlines or completion.
- R5.1.4. Users shall be able to delete goals.

R5.2. AI-Based Assistance

- R5.2.1. AI shall recommend steps based on income/expenses to reach goals.

R5.3. Gamified Reinforcement

- R5.3.1. On success, users earn tier points reflected in AR.
 - R5.3.2. On failure, users lose points and AR assets downgrade accordingly.
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R6. Educational Modules (Subsystem: Learning Engine)

R6.1. Financial Literacy Pathways

- R6.1.1. The system shall provide text-based learning modules and quizzes.
- R6.1.2. Modules shall be grouped into finance topics.

R6.2. Rewards & Incentives

- R6.2.1. Users shall earn points upon successful quiz completion.
 - R6.2.2. Rewards can be used to unlock AR features.
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R7. Personalization (Subsystem: Settings & Customization)

R7.1. AR Environment Customization

- R7.1.1. Users shall customize their AR world.
- R7.1.2. Themes shall be unlocked using in-game points.

R7.2. Notification Preferences

- R7.2.1. Users shall set notification types (reminders, milestones, challenges).
 - R7.2.2. Users shall adjust frequency and delivery method (in-app, email).
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R8. Achievements (Subsystem: Gamification & Progress)

R8.1. Achievement Tracking

- R8.1.1. The system shall track user actions that contribute to achievements (e.g. saving, budgeting, goal completion).
- R8.1.2. Achievements shall be triggered based on predefined rules and thresholds.

R8.2. Achievement Display

- R8.2.1. Each achievement shall include a title, description, and associated reward.
- R8.2.2. Locked achievements shall be displayed with hints or progress indicators.

R8.3. Reward System

- R8.3.1. Achievements shall grant in-game points, badges, or unlockables (e.g. themes, avatar upgrades).
- R8.3.2. Rewards shall be tiered based on achievement difficulty or impact.

R8.4. Progress Feedback

- R8.4.1. The system shall notify users upon unlocking an achievement via in-app alerts or AR animations.

R8.5. Social & Community Integration

- R8.5.1. Users shall optionally share achievements with friends or community groups.
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Subsystems

- Authorization & User Profile
 - 3D/AR Visualization Engine
 - Finance Manager
 - Finance Insights
 - Social Layer
 - Learning Engine
 - Settings and Customization
- 
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Traceability Matrix

	Authenticati on & User Profile	AR Visuali zation Engine	Financ e Manag er	AI Classif icatio n & Insigh ts	Social Layer	Financ ial Goal Engine	Learni ng Engine	Settin gs & Custo mizati on
R1.1	X							
R1.2	X							
R1.3	X							
R2.1		X						
R2.2		X						
R3.1			X	X				
R3.2			X					
R3.3			X					
R3.4			X	X				
R4.1					X			
R4.2					X			



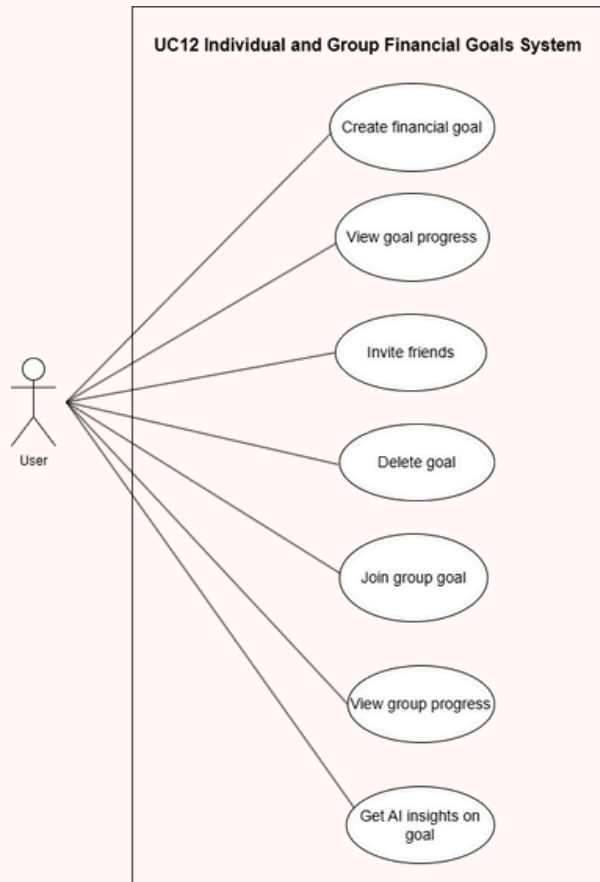
Traceability Matrix

	Authenticati on & User Profile	AR Visuali zation Engine	Financ e Manag er	AI Classif ication & Insigh ts	Social Layer	Financ ial Goal Engine	Learni ng Engine	Settin gs & Custo mizati on
R5.1						X		
R5.2				X		X		
R5.3		X				X		
R6.1							X	
R6.2							X	
R6.3							X	
R7.1								X
R.7.2								X

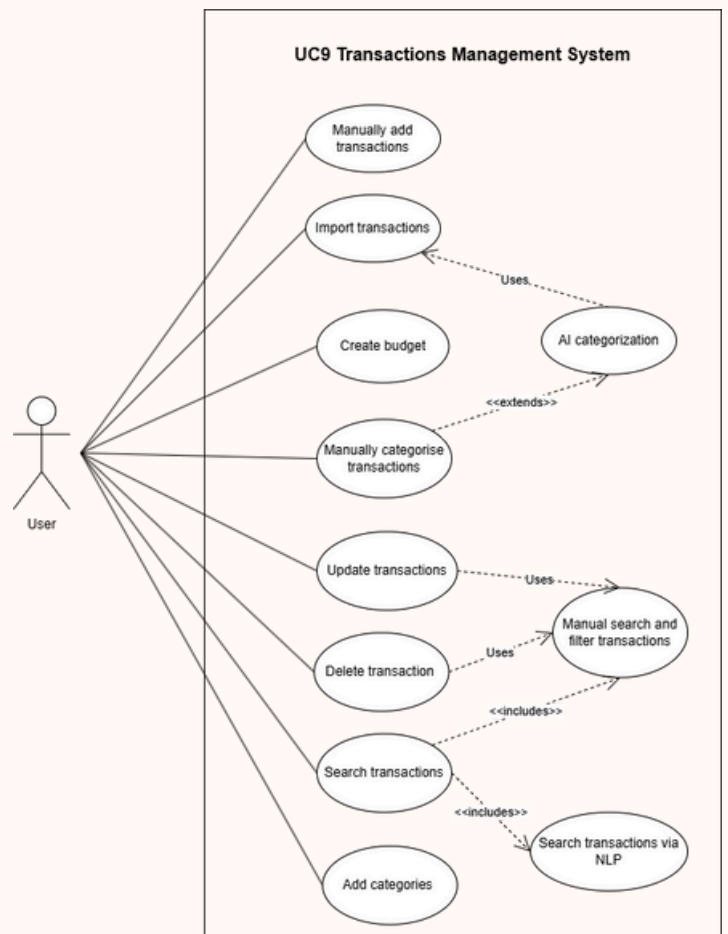


Use Cases

UC12: Financial Goals

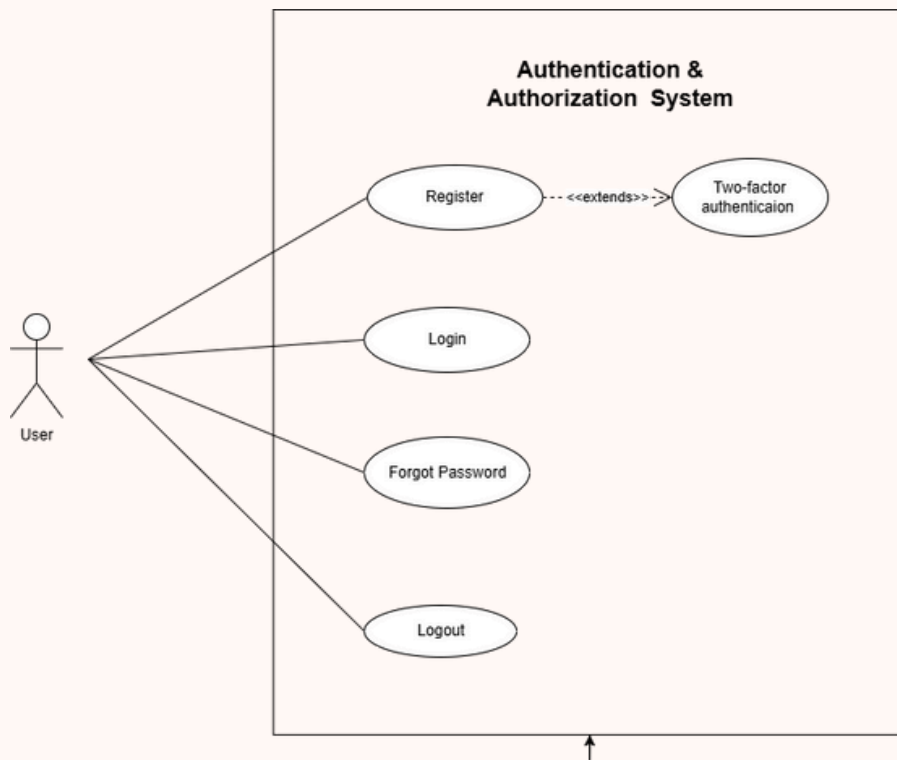


UC9: Transactions

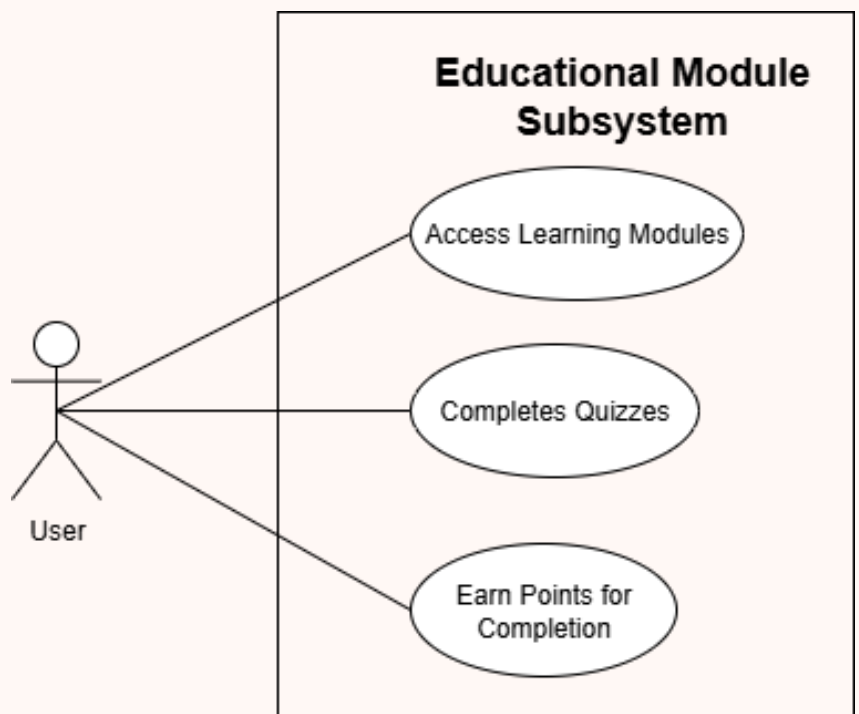


Use Cases

Authentication/Authorization

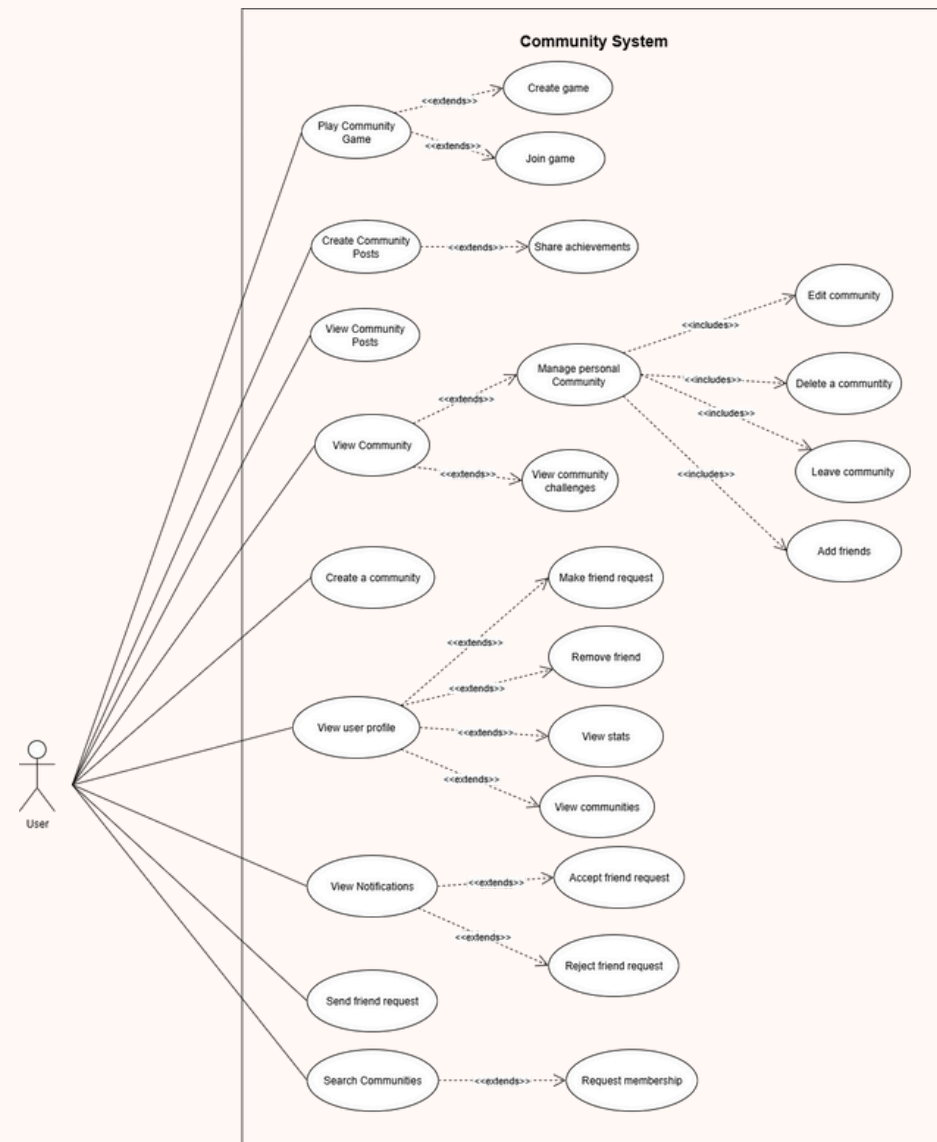


Education

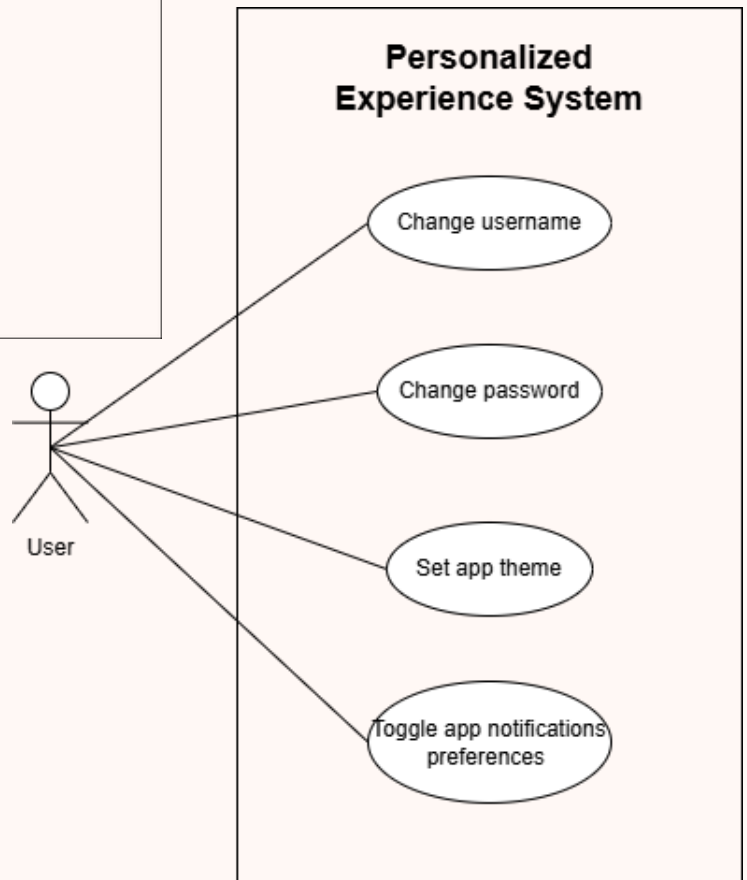


Use Cases

Community

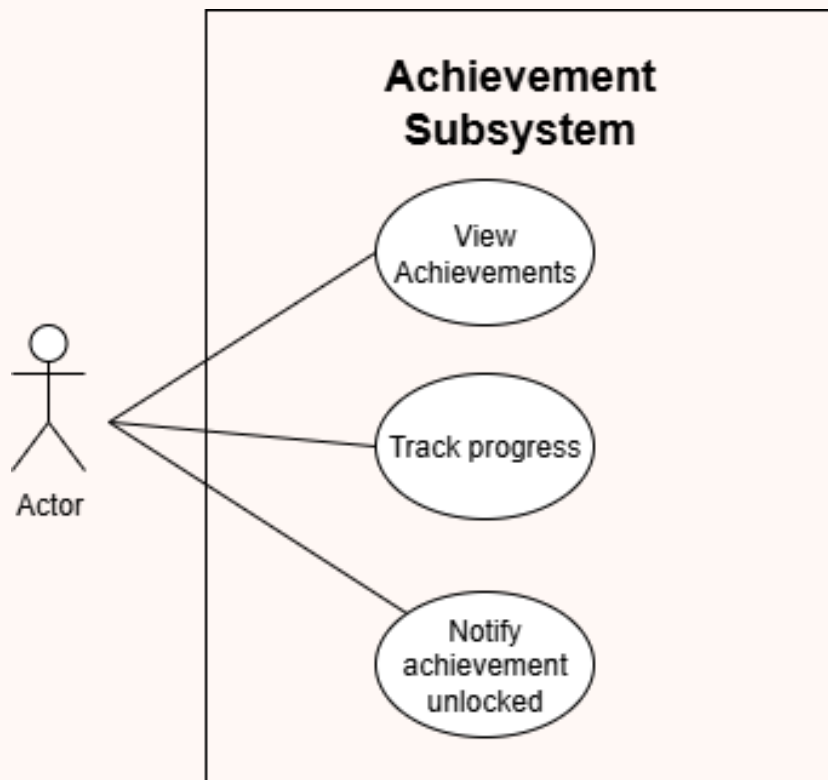


Personalization

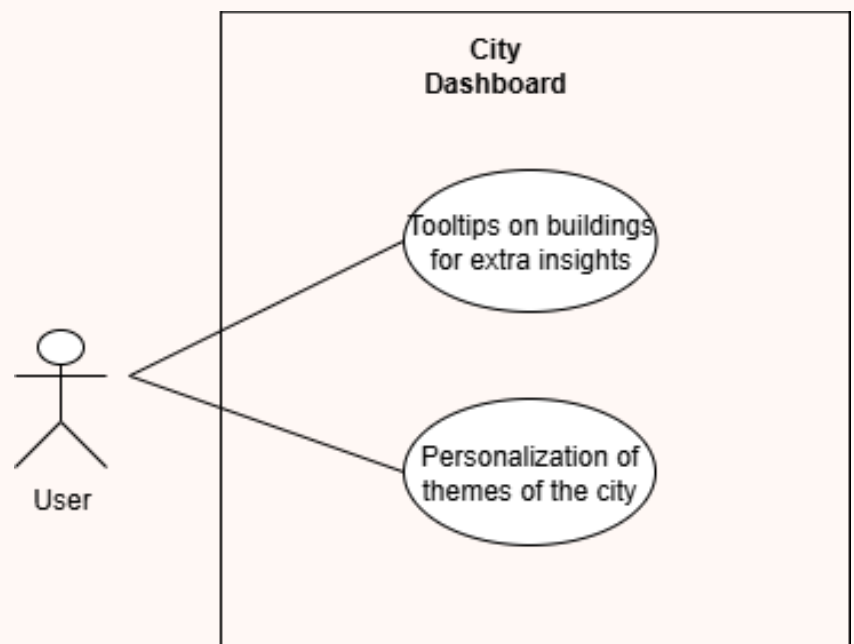


Use Cases

Achievements

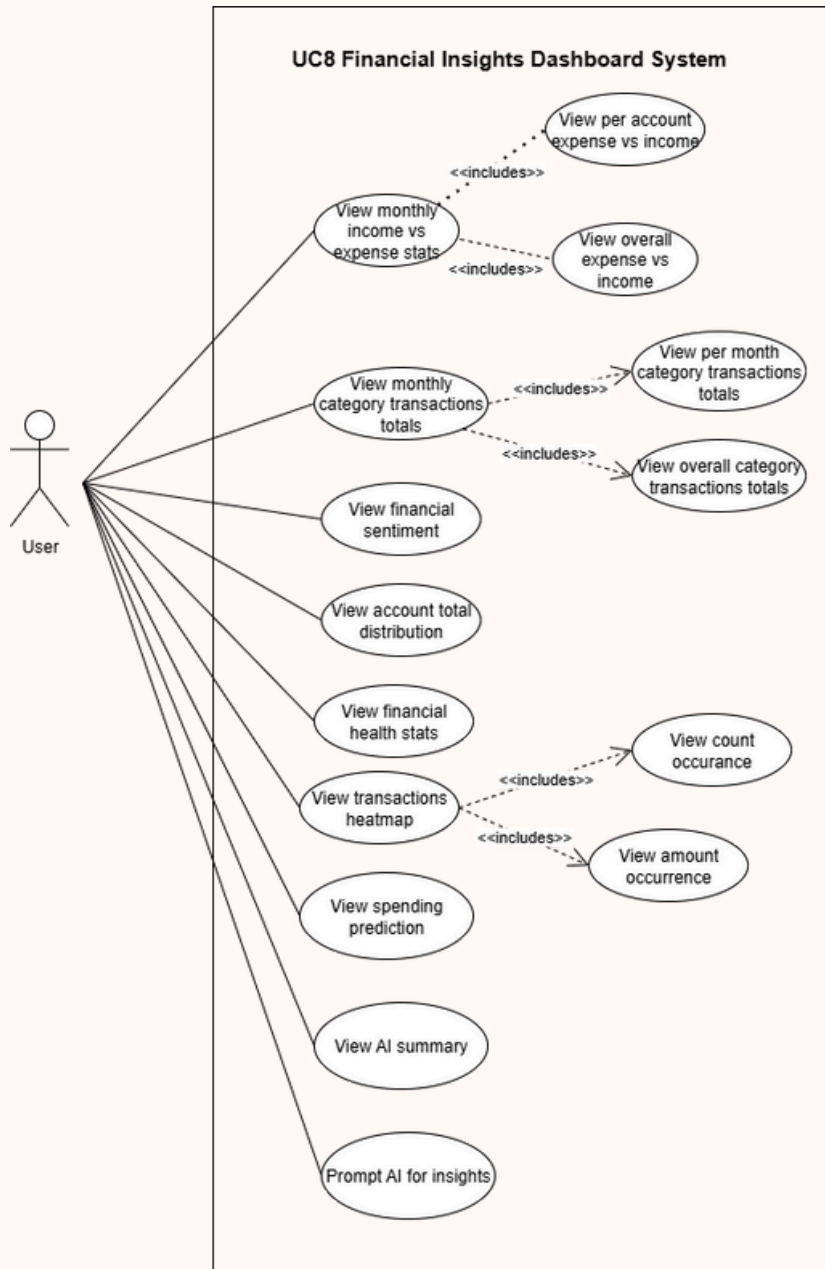


City Dashboard

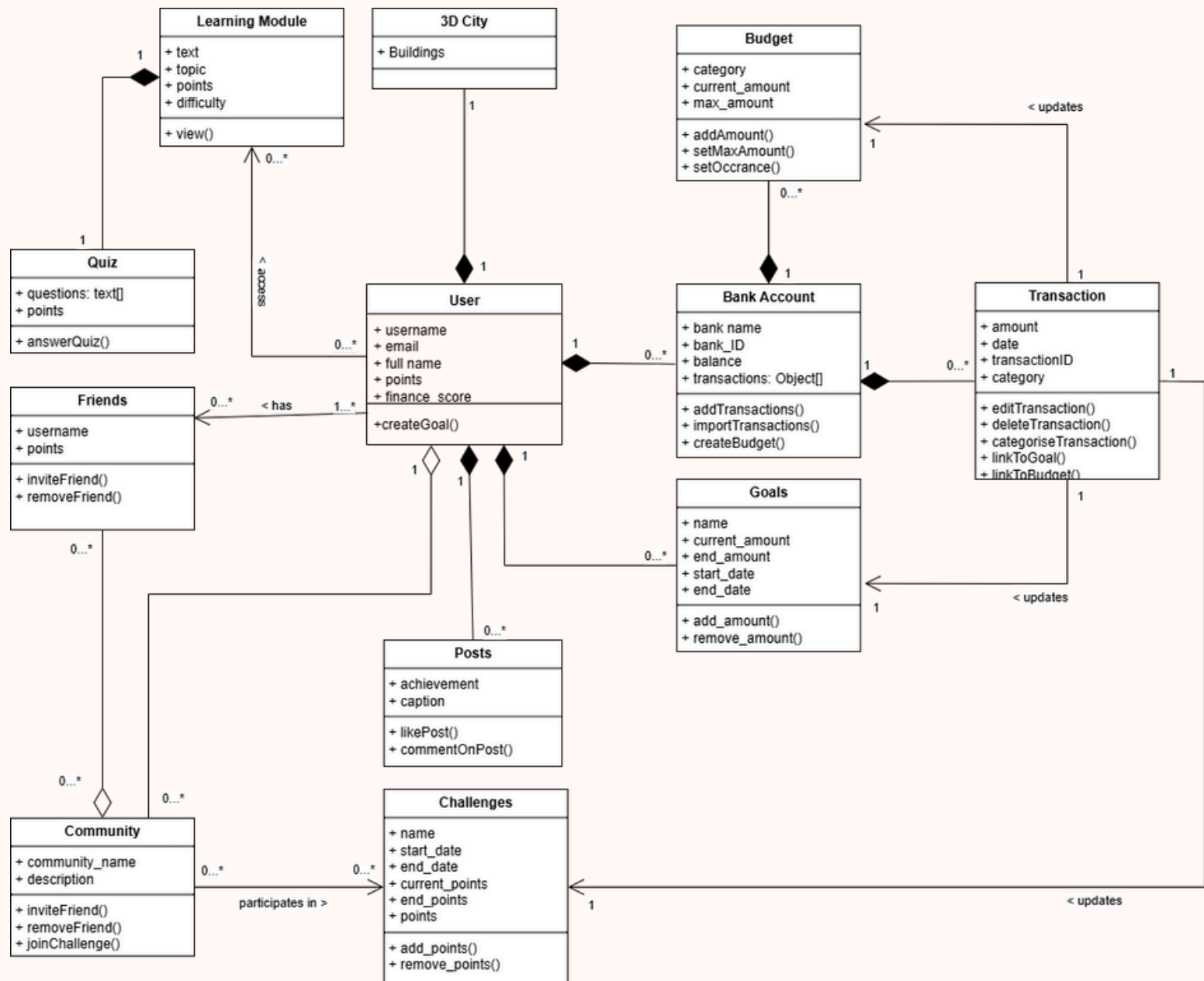


Use Cases

Insights dashboard



Domain Model V3



Service Contract

[External Link to API Specification Document](#)



Gamified Finance
In Collaboration With



CodeBlooded